Do Now · retrieval

Lesson 067 · Divide and Conquer: Splitting the Search

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Define decomposition.

Q2

Which input assumption does binary search require?

Q3

How does Math.Min(a,b) behave?

Q4

Binary search [2,4,6,8,10,12,14] for 12: show low/high and midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define and predict — Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

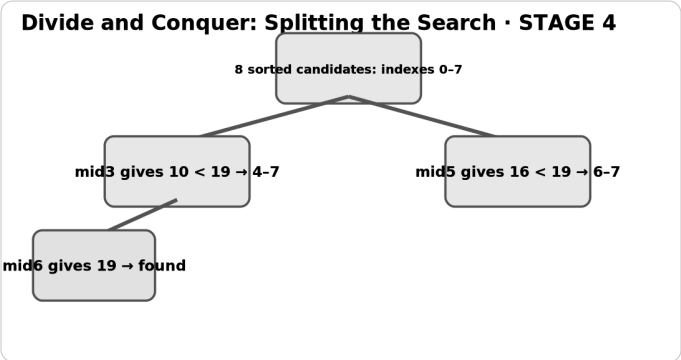
Core idea 2

Explain — Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

Core idea 3

Evaluate — Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

divide and conquer — Solve smaller subproblems and combine.

precondition — Requirement for correctness.

binary search — Halving search on sorted data.

merge — Combine sorted partial results.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

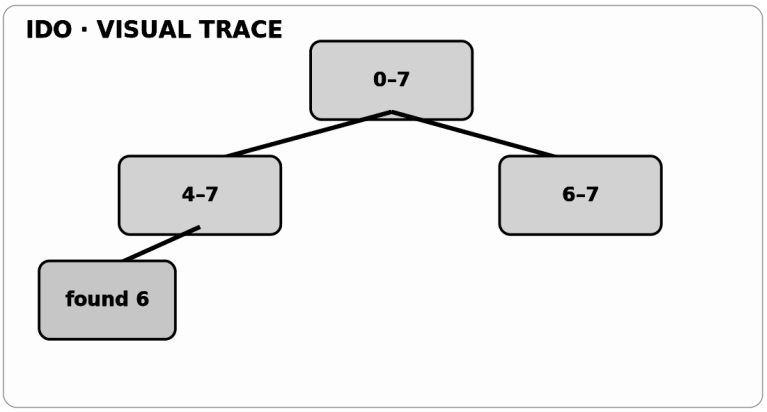
Problem

Search sorted [2,5,8,11,14,17,20,23] for 20.

Follow the worked method

1. Follow the reasoning — Each iteration narrows the candidate interval, and result is index 6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
The second comparison finds the target after removing upper half.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For [1,4,7,10,13,16,19,22] trace middle indexes when target19. [6]

Question 2

Explain divide, solve and combine for merge sort on [7,2,9,1]. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare repeated halving with linear scan for sorted list of 1024 items, in terms of worst-case comparisons and preconditions. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Binary search will correctly find any target in any unsorted list.

Use key vocabulary from memory

1. Explain divide and conquer in your own words.

2. Explain precondition in your own words.

3. Explain binary search in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.